

Unifying Special Relativity and Quantum Mechanics

Lecture 27 — The Coset Space, the Goldstone Matrix, and the Transformation Properties of Goldstone Bosons

($\mathcal{M}_0 \cong G/H$ in detail, the exponential parametrization for a general breaking pattern, the non-linear shift under broken generators and the linear action of H , $\text{Ad}_G \simeq \text{Ad}_H \oplus R_\pi$, and the reintroduction of the Higgs mode)

Typed version of the handwritten notes, with added explanations in blue boxes

Added explanation (not in the original notes) — How to read this document

Conventions. Exactly as in Lectures 1–26: normal text and the yellow / pink / grey boxes are a faithful transcription of the handwritten sheets, and the text colours follow the pens of the original (blue, magenta, green, red, violet, pencil). Everything inside a blue box like this one is material I have added; it is not in the original notes.

Pagination. Still the same chapter, still twelve stars, still no restart. Lecture 24 ran [p. 1*****]–[p. 22*****], Lecture 25 [p. 23*****]–[p. 42*****], Lecture 26 [p. 43*****]–[p. 52*****], and this scan picks up at [p. 53*****] and runs to [p. 72*****]. Twenty sheets, circled numbers consecutive, no duplicates, nothing missing, nothing out of order. PDF page n of this file is sheet $52 + n$.

It picks up exactly where Lecture 26 stopped, and again stops mid-thought. Lecture 26 ended on sheet [p. 52*****] having just introduced the vacuum manifold $\mathcal{M}_0$ and stated the two facts needed to identify it with G/H . Sheet [p. 53*****] opens with the $SO(3) \rightarrow SO(2)$ picture and states $\mathcal{M}_0 \cong G/H$ outright; sheets [p. 54*****]–[p. 57*****] then justify it carefully. At the far end, sheet [p. 72*****] opens a new section, “Goldstone bosons as local coordinates on the vacuum manifold”, sets up the $SO(3) \rightarrow SO(2)$ sphere with its polar angles, and stops at the words “*we start from a simple identity: a generic...*”. Section 14 records the fragment as it stands, and a blue box sketches where it is heading.

One slip and one index-ordering trap flagged in place.

1. On [p. 64*****] the commutator is expanded as $[T^a, X^\alpha] = if_{a\alpha b}T^b + if_{a\alpha\beta}X^\beta$, and then the surviving piece is rewritten as $-if_{a\beta\alpha}X^\beta$. The two differ by the antisymmetry $f_{a\alpha\beta} = -f_{a\beta\alpha}$; nothing is wrong, but the step is silent and is easy to read as a sign error. Spelled out in Section 9.
2. On [p. 67*****] the block-diagonal matrix is written with the *unbroken* indices first (upper-left block = Ad_H), whereas the explicit $SO(3)$ example on [p. 69*****] has the *broken* indices 1, 2 first and the unbroken index 3 last, so the two blocks appear in the opposite order. Both are correct; only the labelling of the basis differs. Noted at the example.

Notation carried over. T^a are unbroken generators ($T^a\Phi_0 = 0$, spanning $\mathfrak{h}$, $a = 1, \dots, \dim H$); X^α are broken ones ($X^\alpha\Phi_0 \neq 0$, $\alpha = 1, \dots, \dim G - \dim H$); T^A runs over both. The key structural fact proved in Lecture 25 [p. 41*****] and used constantly here is $f_{ab\alpha} = 0$.

Added explanation (not in the original notes) — Where we left off — recap of Lecture 26

Lecture 26 completed the proof of Goldstone’s theorem. Evaluating the differentiated invariance condition of the potential at the vacuum gives $M_{ac}^2 (T^A\Phi_0)^a = 0$: for unbroken generators this is empty, for broken ones it says $X^\alpha\Phi_0$ is a non-zero null eigenvector of the mass matrix.

Hence one massless scalar per broken generator, $\# \text{Goldstones} = \dim G - \dim H$.

It then verified this in two families. $SO(N)$ with N real scalars in the fundamental: vacuum manifold S^{N-1} , unbroken $SO(N-1)$, $N-1$ Goldstones. $SU(N)$ with N complex scalars: vacuum manifold S^{2N-1} , unbroken $SU(N-1)$, $2N-1$ Goldstones — of $2N$ real degrees of freedom exactly one, $\text{Re}(\rho_N)$, is massive with $m^2 = 2\mu^2$.

It closed with three comments. (1) The breaking pattern depends on the representation, not just the group. (2) The Cartesian field variables hide the Goldstone structure — the θ_a still appear in the potential — and something like the exponential parametrization of Lecture 24 is wanted. (3) In both examples $\# \text{Goldstones} = \dim \mathcal{M}_0$.

This lecture answers comments (3) and (2), in that order. Sheets [p. 53*****]–[p. 58*****] prove $\mathcal{M}_0 \cong G/H$ and hence $\dim \mathcal{M}_0 = \dim G - \dim H$. Sheets [p. 58*****]–[p. 69*****] construct the general exponential parametrization — the *Goldstone matrix* — and work out how the Goldstone fields transform under both the broken and the unbroken generators. Sheets [p. 70*****]–[p. 72*****] put the Higgs mode back in.

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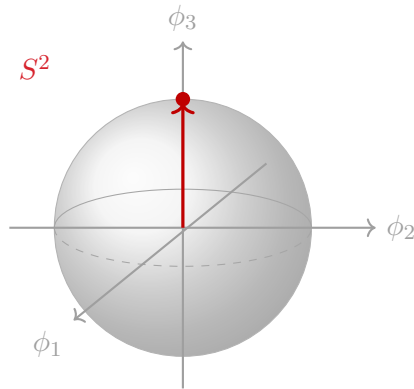
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PART I — The vacuum manifold as a coset space [p. 53*****]— [p. 58*****]

1 Example $SO(3) \rightarrow SO(2)$, and the statement $\mathcal{M}_0 \cong G/H$ [p. 53*****]

EXAMPLE $SO(3) \rightarrow SO(2)$



- The vacuum manifold is the **two-sphere**.
- Points on S^2 are related by $SO(3)$ transformations.

We now choose a vacuum configuration. For example

$$\Phi_0 = \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix}$$

- Φ_0 is invariant for rotations around the third axis, $SO(2)$.

These considerations can be better formalized by saying that

$$\mathcal{M}_0 \cong G/H$$

The vacuum manifold is isomorphic to the quotient space (or coset space) $G \bmod H$.

Obs. Since H is not normal, G/H is not a subgroup.

Added explanation (not in the original notes)

The example in numbers. $\dim SO(3) = 3$, $\dim SO(2) = 1$, so $\dim G - \dim H = 2$ — two Goldstone bosons — and indeed $\dim S^2 = 2$. This is the $N = 3$ case of the family $SO(N) \rightarrow SO(N-1)$ from Lecture 26 [p. 44*****]–[p. 47*****], and it is the smallest one in which the unbroken subgroup is non-trivial, which is why it is the running example for the rest of the lecture.

On the “Obs.” A quotient G/H inherits a group structure only when H is *normal*, i.e. $gHg^{-1} = H$ for all g . Lecture 25 [p. 41*****] showed that for a spontaneously broken symmetry H is *not* normal — the commutator $[X^\alpha, T^a]$ points along broken directions, out of $\mathfrak{h}$. So G/H is a *manifold* (a homogeneous space) and nothing more: its points are cosets, and there is no natural way to multiply two of them. That is fine — the vacuum manifold is not a group either. Concretely, S^2 carries no group structure, whereas $S^1 \cong U(1)$ (Lecture 24) does, precisely because there H was trivial and hence normal.

Why the statement is not obvious. $\mathcal{M}_0$ is defined by a condition on the *potential* ($V = V_{\min}$); G/H is defined by *group theory* alone. That the two coincide is the content of [p. 54*****]–[p. 57*****], and it is what makes the counting of Goldstone bosons a purely group-theoretic question — no knowledge of the potential required beyond which subgroup survives.

2 Reminder: the construction of the coset space G/H [p. 54*****]— [p. 55*****]

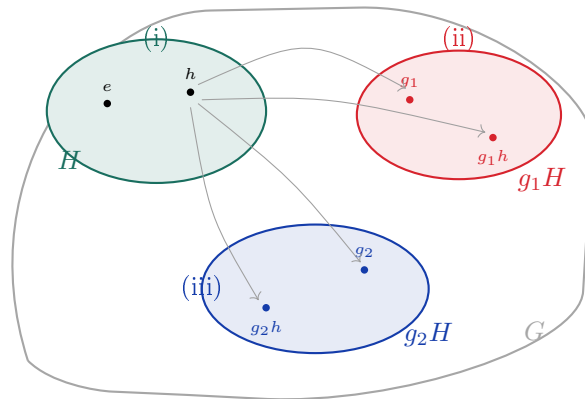
Let us discuss this correspondence a bit more in detail. The key aspects are the following.

1) First, we remind the construction of the coset space G/H .

We organize the elements of G in the so-called “left cosets”.

i) We consider the elements of the subgroup H

ii) Then, we consider $g_1 \notin H$ and we construct the set



$$g_1 H \equiv \{g_1 h \mid h \in H\}$$

iii) Then, we consider $g_2 \notin H$, $g_2 \notin g_1 H$ and construct

$$g_2 H \equiv \{g_2 h \mid h \in H\}$$

We continue until we exhaust the group elements.

The set of all left cosets $\{H, g_1 H, g_2 H, \dots\}$ is called the **coset space**.

Added explanation (not in the original notes)

Why the cosets partition G . Every $g \in G$ lies in the coset gH (take $h = e$), so the cosets cover G . And two cosets are either identical or disjoint: if $g_1 h_1 = g_2 h_2$ for some $h_1, h_2 \in H$, then $g_2^{-1} g_1 = h_2 h_1^{-1} \in H$, so $g_1 \in g_2 H$ and therefore $g_1 H = g_2 H$. This is exactly why the recipe of the notes terminates and produces a clean partition of G into non-overlapping blobs, as drawn.

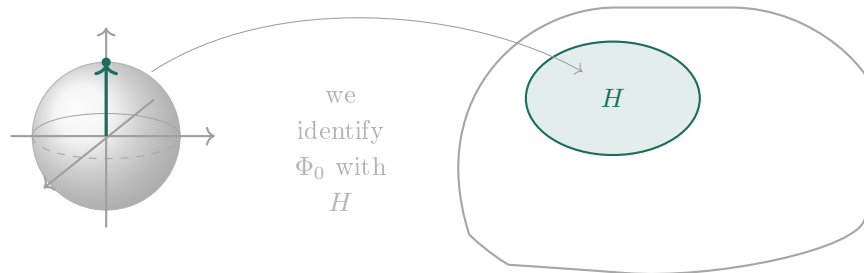
All cosets have the same size. The map $h \mapsto gh$ is a bijection from H to gH (its inverse is left multiplication by g^{-1}). So every blob is a faithful copy of H — which is why, for finite groups, $|G| = |G/H| \cdot |H|$ (Lagrange’s theorem), and for Lie groups $\dim G = \dim(G/H) + \dim H$. The dimension formula on [p. 58*****] is the continuous version of exactly this statement.

Left versus right. One could equally build *right* cosets Hg . When H is normal the two coincide and G/H is a group; when it is not — our case — they differ, and one must choose. The convention here is left cosets, which is what matches $\Phi = g\Phi_0$ with g acting on the left.

3 The correspondence between $\mathcal{M}_0$ and G/H [p. 55*****]— [p. 56*****]

2) Next, we consider $\mathcal{M}_0$.

i) We start from a specific field configuration $\Phi_0 \in \mathcal{M}_0$. By definition we have $H\Phi_0 = \Phi_0$ for all $h \in H$. We thus have a correspondence between Φ_0 and H .



ii) Next, we consider $g_1 \notin H$ and consider $g_1 H \Phi_0$. We find

$$g_1 H \Phi_0 = g_1 (H \Phi_0) = g_1 \Phi_0 = \Phi_1 \neq \Phi_0$$

▷ The action of $g_1 \notin H$ on Φ_0 creates another vacuum different from Φ_0 but still in $\mathcal{M}_0$.

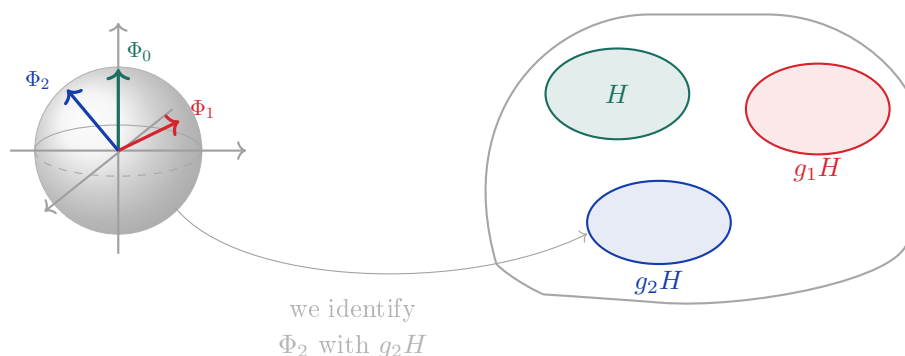
This is true whatever $h \in H$: $g_1 H \Phi_0 = \Phi_1$ is always equal to Φ_1 . We thus identify Φ_1 with $g_1 H$.

iii) We consider $g_2 \notin H$, $g_2 \notin g_1 H$, and compute $g_2 H \Phi_0$. We find

$$g_2 H \Phi_0 = g_2 \Phi_0 = \Phi_2 \neq \Phi_0 \quad (\Phi_2 \neq \Phi_1)$$

▷ We get another vacuum configuration that belongs to $\mathcal{M}_0$, but different from Φ_0 and Φ_1 .

Since $g_2 H \Phi_0 = \Phi_2 \forall h \in H$, we identify Φ_2 with $g_2 H$.



Added explanation (not in the original notes)

Why the whole coset gives one and the same vacuum. The computation $g_1 H \Phi_0 = g_1 (H \Phi_0) = g_1 \Phi_0$ uses only the defining property $h \Phi_0 = \Phi_0$. So every element of the coset $g_1 H$ — an entire $\dim H$ -parameter family of group elements — maps Φ_0 to the *same* point Φ_1 . The coset is precisely the set of group elements that cannot be distinguished by their action on the vacuum. That is the content of the identification, and it is why the redundancy

counted on [p. 57*****]–[p. 58*****] is exactly $\dim H$.

Reading the two pictures. On the left, three points on the sphere: the north pole Φ_0 and two other vacua reached by rotating it. On the right, the group $SO(3)$ carved into blobs. The correspondence is one blob $\leftrightarrow$ one point of the sphere. Filling in all the blobs fills in the whole sphere — which is the assumption (transitivity) stated on [p. 52*****].

4 The map is bijective; the dimension of G/H [p. 57*****]–[p. 58*****]

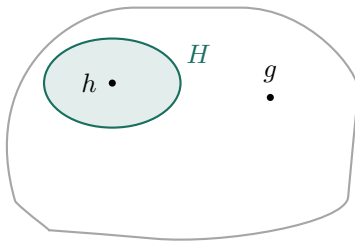
The map constructed in this way is bijective and represents the isomorphism that establishes the relation

$$\mathcal{M}_0 \cong G/H$$

We now ask, what is the dimension of G/H ?

$$\dim(\mathcal{M}_0) = \dim(G/H) = ?$$

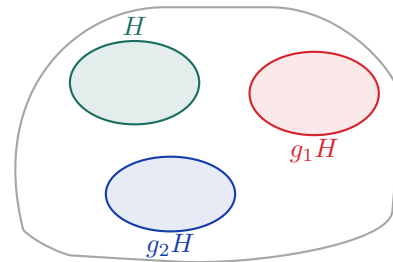
We answer this question with the following intuition.



← To specify a generic element $g \in G$, we need $\dim G$ parameters.

← To specify a generic element $h \in H$ of the subgroup H , we need $\dim H$ parameters.

If we think in terms of G/H , we “group together” elements of G that differ by H transformations.



$\leftrightarrow$ Consequently, if we want to point to an element of the coset space, we do not really need to specify what $h \in H$ since we will always end up in the same coset gH .

$\rightsquigarrow$ In other words, $\dim H$ parameters are redundant.

Consequently, to specify an element of the coset space, we only need to specify $\dim G - \dim H$ parameters.

We find

$$\dim \mathcal{M}_0 = \dim(G/H) = \dim G - \dim H$$

$\Downarrow$ since $\dim G - \dim H = \# \text{ Goldstone bosons}$

$$\# \begin{matrix} \text{Goldstone} \\ \text{bosons} \end{matrix} = \dim \mathcal{M}_0$$

Added explanation (not in the original notes)

This closes comment 3 of Lecture 26. The observation there — that in both worked examples the number of Goldstones equalled the dimension of the vacuum manifold — is now explained rather than noticed. The chain is: one Goldstone per broken generator (Goldstone's theorem, Lecture 26 [p. 43*****]); broken generators = $\dim G - \dim H$; and $\dim \mathcal{M}_0 = \dim(G/H) = \dim G - \dim H$ by the counting above.

The counting argument, stated precisely. Locally, near the identity, G is parametrized by $\dim G$ real coordinates and H by $\dim H$ of them. The quotient map $G \rightarrow G/H$ has fibres that are copies of H , so it collapses $\dim H$ directions. A smooth surjection whose fibres are $\dim H$ -dimensional lowers the dimension by exactly that much — hence $\dim(G/H) = \dim G - \dim H$. (The word “intuition” in the notes is well chosen: the rigorous version requires knowing that G/H is a smooth manifold and the projection a submersion, which holds for H a closed subgroup of a Lie group.)

Check on the examples. $SO(3)/SO(2)$: $3 - 1 = 2 = \dim S^2$. ✓ $SO(N)/SO(N-1)$: $\frac{N(N-1)}{2} - \frac{(N-1)(N-2)}{2} = N-1 = \dim S^{N-1}$. ✓ $SU(N)/SU(N-1)$: $(N^2-1) - ((N-1)^2-1) = 2N-1 = \dim S^{2N-1}$. ✓

PART II — The Goldstone matrix [p. 58*****]–[p. 62*****]

5 Looking for a better parametrization [p. 58*****]–[p. 59*****]

■ **The Goldstone matrix** (point 2)

We now look for a better parametrization of the Goldstone fields.

↪ Something analogous to the exponential parametrization in the case of $SO(2)$.

Reminder: writing $\varrho(x) = \frac{1}{\sqrt{2}}(v + h(x))e^{i\pi(x)/f_\pi}$ was useful because in the combination $\varrho^*\varrho$ the presence of $\pi(x)$ cancels out $\rightsquigarrow$ there was no dependence on $\pi(x)$ in the potential.

↪ We look for something similar: a field parametrization such that $\pi(x)$ completely disappears from the potential.

We put forward the following **ansatz**:

$$\Phi(x) = \exp\left(\frac{i\pi^A(x)T^A}{f_\pi}\right)\Phi_0$$

Φ_0 is the vacuum after SSB; for the moment we do not include the Higgs mode.

This seems to be a good ansatz. We take our Lagrangian, $\mathcal{L} = \mathcal{L}_{\text{KIN}} - V(\Phi)$, and plug in the above parametrization. The potential becomes

$$V(\pi) = V\left(\exp\left(\frac{i\pi^A(x)T^A}{f_\pi}\right)\Phi_0\right)$$

↪ The crucial observation is that, $\forall x$ spacetime point, $\exp(i\pi^A(x)T^A/f_\pi) \in G$ is an element of the group.

Added explanation (not in the original notes)

The idea in one sentence. Instead of writing the field as “vacuum + fluctuation” in Cartesian coordinates, write it as “a group transformation applied to the vacuum”. Since the potential is G -invariant and Φ_0 is a minimum, the transformed configuration is *still* a minimum — so the potential cannot depend on the transformation parameters at all. Those parameters are the Goldstone fields.

Why the exponential. Any group element continuously connected to the identity can be written $\exp(i\epsilon^A T^A)$. Promoting the constants ϵ^A to spacetime-dependent fields $\pi^A(x)/f_\pi$ is the generalization of Lecture 24’s $e^{i\pi(x)/v}$, where the one-parameter group $U(1)$ had a single generator. As there, f_π is a constant of mass dimension 1 inserted so that $\pi^A(x)$ is a canonically normalizable field.

Note what is deliberately missing. The ansatz as written has no radial direction: the configuration $\Phi(x)$ always lies exactly *on* the vacuum manifold, for every x . That is why the potential will come out constant. The massive Higgs mode — the fluctuation *off* the manifold — is put back in on [p. 70*****]. Keeping them separate is the right order: first show the Goldstones are pure gauge with respect to V , then add the one direction that is not.

6 The potential is constant; restriction to broken directions

[p. 60*****]

Consequently, we see that

$$\exp\left(\frac{i\pi^A(x)T^A}{f_\pi}\right)\Phi_0 = \Phi'_0 \in \mathcal{M}_0$$

In general, $\Phi'_0 \neq \Phi_0$, but Φ'_0 is still a vacuum configuration that minimizes the energy.

We thus find:

$$V(\pi) = V\left(\exp\left(\frac{i\pi^A(x)T^A}{f_\pi}\right)\Phi_0\right) = V(\Phi'_0) = V_{\min}$$

$\hookrightarrow$ The presence of the Goldstone fields completely disappears from the potential, exactly as we wanted.

There is still something to adjust. If we sum over $A = 1, \dots, \dim G$, we will have too many Goldstones, since we only have $\dim G - \dim H$ of them.

$\rightsquigarrow$ The point is that, if we consider a combination of unbroken generators, we have

$$\Phi(x) = \exp\left(\frac{i\pi^a(x)T^a}{f_\pi}\right)\Phi_0 = \Phi_0$$

so that all the unbroken directions are redundant.

Consequently, we restrict to broken directions only, and write:

$$\Phi(x) = \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right)\Phi_0$$

a.k.a. Goldstone matrix

Added explanation (not in the original notes)

The redundancy, spelled out. If T^a is unbroken then $T^a\Phi_0 = 0$, hence $(T^a)^n\Phi_0 = 0$ for all $n \geq 1$, and the exponential series collapses: $\exp(i\epsilon_a T^a)\Phi_0 = \Phi_0 + i\epsilon_a T^a\Phi_0 + \dots = \Phi_0$. So the unbroken parameters do nothing whatsoever — they are precisely the $\dim H$ redundant coordinates identified on [p. 57*****]–[p. 58*****]. Restricting the sum to X^α leaves exactly $\dim G - \dim H$ fields, matching the Goldstone count.

This is the same counting seen for the third time. $\dim H$ redundant parameters = $\dim H$ directions collapsed by the quotient $G/H = \dim H$ unbroken generators annihilating the vacuum. The Goldstone matrix is a concrete coordinate system on the coset space, and the π^α are literally coordinates on $\mathcal{M}_0$ — which is the title of the section the lecture stops in the middle of ([p. 72*****]).

One technical caveat. The map $\pi^\alpha \mapsto \exp(i\pi^\alpha X^\alpha/f_\pi)H$ is a bijection onto the coset space only *locally*, near $\pi = 0$; globally a single exponential chart cannot cover a compact manifold like S^2 (the exponential map wraps and degenerates). This is harmless for perturbation theory, which only ever expands around $\pi = 0$, but it is the reason for the word “local” in “local coordinates on the vacuum manifold”.

And a caveat about the ordering. Because the broken generators do *not* close (Lecture 25 [p. 41*****]), the product of two Goldstone matrices is not a Goldstone matrix: in general $e^{iaX}e^{ibX} = e^{icX}e^{i\epsilon T}$ with a leftover H element. That leftover is exactly what makes the transformation law of [p. 61*****]–[p. 65*****] non-trivial, and it is the technical heart of the CCWZ construction.

7 Transformations generated by broken symmetries [p. 61*****]–[p. 62*****]

■ Transformation properties of the Goldstone fields

→ Transformations generated by broken symmetries

We consider

$$\Phi(x) = \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \Phi_0 \longrightarrow \exp(-i\theta_\alpha X^\alpha) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \Phi_0$$

★

It is not easy to work out this transformation in general.

We consider a transformation in which we only exponentiate a real linear combination of broken generators.

We thus opt for a “double expansion” in powers of both the parameters θ_α and the Goldstone fields.

Left-hand side of ★: we expand

$$\Phi(x) = \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \Phi_0 = \left[\mathbb{1} + \frac{i\pi^\alpha X^\alpha}{f_\pi} + \mathcal{O}\left(\frac{\pi^2}{f_\pi^2}\right)\right] \Phi_0$$

Right-hand side of ★: we write

$$\begin{aligned} \exp(-i\theta_\alpha X^\alpha) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \Phi_0 &= \\ &= (\mathbb{1} - i\theta_\alpha X^\alpha + \mathcal{O}(\theta^2)) \left(\mathbb{1} + \frac{i\pi^\alpha X^\alpha}{f_\pi} + \mathcal{O}\left(\frac{\pi^2}{f_\pi^2}\right)\right) \Phi_0 = \\ &= \left[\mathbb{1} - i\theta_\alpha X^\alpha + \frac{i\pi^\alpha X^\alpha}{f_\pi} + \dots\right] \Phi_0 = \\ &= \left[\mathbb{1} + i\left(\frac{\pi^\alpha}{f_\pi} - \theta_\alpha\right) X^\alpha + \dots\right] \Phi_0 \end{aligned}$$

From the comparison, we write

$$\pi^\alpha(x) \longrightarrow \pi^\alpha(x) - f_\pi \theta_\alpha$$

Non-linear shift under the broken symmetry

Under the broken symmetry, Goldstone bosons transform non-linearly.

↔ It is this non-linearly realized symmetry that forbids non-derivative terms in the Goldstone boson Lagrangian.

Added explanation (not in the original notes)

This is Lecture 24 [p. 20***]–[p. 21*****] in general form.** There, with a single broken generator, the transformation was $\pi \rightarrow \pi + c$ — an exact shift with no corrections. Here the same thing happens for each broken direction: $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$, at leading order in the double expansion. The shift is *field-independent*, hence non-linear in the technical sense of Lecture 24 [p. 20*****]: the transformation does not vanish at $\pi = 0$, so the vacuum is not a fixed point.

Why only to leading order. In the abelian case the shift was exact because $e^{-i\theta X} e^{i\pi X/f} = e^{i(\pi/f - \theta)X}$ — the exponents simply add. With several broken generators they do not: the Baker–Campbell–Hausdorff formula produces $-\frac{1}{2}[\theta_\alpha X^\alpha, \pi^\beta X^\beta/f_\pi]$ and higher terms, and $[X, X] \sim T + X$ can land partly in $\mathfrak{h}$. The corrections are field-dependent and are what the notes mean by “not easy to work out in general”. To the order needed — one power of θ , one of π — the simple shift is exact, and that is enough to establish the symmetry.

Why the shift forbids non-derivative terms. Any candidate term in $\mathcal{L}$ built from π^α without derivatives — a mass term $m^2 \pi^\alpha \pi^\alpha$, a potential $V(\pi)$, anything — would change under $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$, since θ_α is arbitrary. Only $\partial_\mu \pi^\alpha$ is invariant, because $\partial_\mu(\text{const}) = 0$. Hence: Goldstones are exactly massless, and every interaction they have carries at least one derivative, to all orders. This is the symmetry-based proof of what the explicit computation of [p. 60*****] showed for the potential.

PART III — Transformations under the unbroken group

[p. 63*****]–[p. 69*****]

8 Rewriting the H -transformation [p. 63*****]

→ Transformations generated by unbroken symmetries

We now investigate

$$\Phi(x) = \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \Phi_0 \longrightarrow \exp(-i\alpha_a T^a) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \Phi_0$$

↔ Let's rewrite this term as follows

$$\begin{aligned} \exp(-i\alpha_a T^a) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \exp(i\alpha_a T^a) & \underbrace{\exp(-i\alpha_a T^a) \Phi_0}_{\text{this is } \Phi_0, \text{ since } T^a \text{ are unbroken}} \\ & = \exp(-i\alpha_a T^a) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \exp(i\alpha_a T^a) \Phi_0 \end{aligned}$$

We can thus rewrite the above transformation rule at the level of the Goldstone matrix as follows

$$\exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \longrightarrow \exp(-i\alpha_a T^a) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \exp(i\alpha_a T^a)$$

Let's elaborate the right-hand side.

We use $Ce^XC^{-1} = e^{CXC^{-1}}$

Added explanation (not in the original notes)

The trick, and why it is the right one. Inserting $\mathbb{1} = e^{i\alpha_a T^a} e^{-i\alpha_a T^a}$ costs nothing, and the rightmost factor then acts on Φ_0 and does nothing, because $T^a \Phi_0 = 0$. What is left is a *conjugation* of the Goldstone matrix. The payoff is that a conjugation acts on the *generators* rather than on the vacuum, so the transformation can be read off entirely at the level of the exponent — and, as the next sheets show, it turns out to be linear in the π^α .

The identity $Ce^XC^{-1} = e^{CXC^{-1}}$. Expand: $Ce^XC^{-1} = \sum_n \frac{1}{n!} CX^n C^{-1}$, and $CX^n C^{-1} = (CXC^{-1})^n$ because the intermediate $C^{-1}C$ pairs cancel. So the conjugation passes through the exponential and acts on X alone. This is used constantly in Lie theory and is worth having at your fingertips.

Why the answer must be linear. H is realized à la Wigner–Weyl — the vacuum is invariant under it — and Lecture 24 [p. 22*****] established that unbroken symmetries act *linearly* on the fields. The computation below simply confirms this and identifies *which* representation of H the Goldstones fall into.

9 The adjoint action on the broken generators [p. 64*****]–

[p. 65*****]

$$\longrightarrow \exp\left[\frac{i\pi^\alpha}{f_\pi} \underbrace{e^{-i\alpha_a T^a} X^\alpha e^{i\alpha_a T^a}}_{(\star)}\right] \quad (\star)$$

We now focus on this term. We expand in small α .

$$\begin{aligned} e^{-i\alpha_a T^a} X^\alpha e^{i\alpha_a T^a} &= (\mathbb{1} - i\alpha_a T^a) X^\alpha (\mathbb{1} + i\alpha_a T^a) = \\ &= X^\alpha - i\alpha_a \underbrace{[T^a, X^\alpha]}_{=0} = X^\alpha - i\alpha_a (if_{a\alpha\beta}) X^\beta \end{aligned}$$

$$\begin{aligned} [T^a, X^\alpha] &= i \underbrace{f_{a\alpha b}}_{=0} T^b + if_{a\alpha\beta} X^\beta \\ &= X^\alpha - i\alpha_a (-if_{a\beta\alpha}) X^\beta = \end{aligned}$$

We define

$$(t_\pi^a)_{\beta\alpha} \equiv -if_{a\beta\alpha}$$

$$= X^\alpha - i\alpha_a (t_\pi^a)_{\beta\alpha} X^\beta$$

We thus get, at the infinitesimal level

$$e^{-i\alpha_a T^a} X^\alpha e^{i\alpha_a T^a} = X^\alpha - i\alpha_a (t_\pi^a)_{\beta\alpha} X^\beta$$

This is the first term in an exponential expansion, and we write

$$e^{-i\alpha_a T^a} X^\alpha e^{i\alpha_a T^a} = \left[\exp(-i\alpha_a t_\pi^a) \right]_{\beta\alpha} X^\beta$$

Added explanation (not in the original notes)

The silent step flagged. Between the two lines the notes replace $if_{a\alpha\beta}$ by $-if_{a\beta\alpha}$. This is just total antisymmetry of the structure constants of a compact group in their last two (indeed all) indices: $f_{a\alpha\beta} = -f_{a\beta\alpha}$. It is done so that the free index α sits in the *second* slot of the matrix $(t_\pi^a)_{\beta\alpha}$, matching the standard convention for a generator acting on a column vector labelled by β . No sign has been lost.

Why the T^b piece drops. $f_{a\alpha b} = 0$ is the result proved in Lecture 25 [p. 41*****]: the commutator of an unbroken with a broken generator has no component along the unbroken directions. Equivalently $[\mathfrak{h}, \mathfrak{g}/\mathfrak{h}] \subset \mathfrak{g}/\mathfrak{h}$. That is exactly the statement that the broken generators form a *representation* of H — and (t_π^a) are its generator matrices.

Why exponentiating the infinitesimal result is legitimate. The map $X \mapsto CXC^{-1}$ with $C = e^{-i\alpha T}$ is a linear map on the space spanned by the X^α , and composing two such conjugations composes the maps. A one-parameter family of linear maps with the given infinitesimal generator is the exponential of that generator — hence the finite form. Checking the algebra: the matrices t_π^a satisfy $[t_\pi^a, t_\pi^b] = if_{abc} t_\pi^c$, inherited from the Jacobi identity, confirming they really are a representation of $\mathfrak{h}$.

10 The Goldstones transform linearly under H [p. 65*****]

Going back to (★), we write

$$\begin{aligned}\exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) &\longrightarrow \exp\left[\frac{i\pi^\alpha}{f_\pi}\left(e^{-i\alpha_a t_\pi^a}\right)_{\beta\alpha} X^\beta\right] = \\ &= \exp\left[\frac{i}{f_\pi}\left(e^{-i\alpha_a t_\pi^a}\right)_{\alpha\beta} \pi^\beta X^\alpha\right]\end{aligned}$$

From the argument of the exponential, we get

$$\pi^\alpha(x) \xrightarrow{H} \left[\exp(-i\alpha_a t_\pi^a)\right]_{\alpha\beta} \pi^\beta(x)$$

with

$$(t_\pi^a)_{\alpha\beta} \equiv -if_{a\alpha\beta}$$

We find that the Goldstone bosons transform linearly under the unbroken group H . This is expected, since H is realized à la Wigner–Weyl.

We can take one step further, and ask: what is the representation of H under which the Goldstone bosons transform?

Added explanation (not in the original notes)

The relabelling in the second line. Going from $\pi^\alpha(e^{-i\alpha t_\pi})_{\beta\alpha} X^\beta$ to $(e^{-i\alpha t_\pi})_{\alpha\beta} \pi^\beta X^\alpha$ is nothing but swapping the dummy names $\alpha \leftrightarrow \beta$. It is done so that the coefficient of X^α can be read directly as the new π^α/f_π , which is what the boxed transformation law states.

Contrast the two transformation laws, side by side ([p. 62*****] versus [p. 65*****]). This is the payoff of the whole section:

broken generators X^α	unbroken generators T^a
$\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$	$\pi^\alpha \rightarrow [e^{-i\alpha_a t_\pi^a}]_{\alpha\beta} \pi^\beta$
non-linear (a shift)	linear (a rotation)
no fixed point: $\pi = 0$ moves	$\pi = 0$ is fixed
forbids non-derivative terms	organizes them into multiplets

Both are symmetries of the same Lagrangian. G is not broken as a symmetry of $\mathcal{L}$ — it is broken only by the vacuum — and the two halves of G simply act on the fields in two different ways.

Why the π^α must fall into a genuine representation. $\pi = 0$ is the chosen vacuum, which H leaves invariant; so H maps small fluctuations to small fluctuations linearly, and linear actions of a group are representations. The only question left — answered next — is *which* one.

11 The adjoint of G restricted to H [p. 66*****]–[p. 68*****]

Goldstone bosons transform?

We found an explicit form of the generators, $(t_\pi^a)_{\alpha\beta} = -if_{a\alpha\beta}$.

$\hookrightarrow$ This form of the generators suggests to look at the adjoint representation.

i) We start by looking at the adjoint representation of G :

$$(t^A)_{BC} = -if_{ABC}$$

with $A, B, C = 1, \dots, \dim G$. We have $\dim G$ matrices (for $A = 1, \dots, \dim G$) and each of them is $\dim G$ -dimensional ($B, C = 1, \dots, \dim G$).

ii) We now restrict our attention to the subgroup H . It means that, instead of taking $A = 1, \dots, \dim G$, we restrict to $A \rightarrow a = 1, \dots, \dim H$.

$$(t^a)_{BC} = -if_{aBC}$$

For each $a = 1, \dots, \dim H$ we still have a $\dim G$ -dimensional matrix (since $B, C = 1, \dots, \dim G$).

↔ Let's inspect a bit more explicitly the structure of these matrices.

$$(t^a)_{BC} = \left(\begin{array}{c|c} -if_{abc} & -if_{ab\alpha} \\ \hline -if_{a\alpha c} & -if_{a\alpha\beta} \end{array} \right)$$

We know that $f_{ab\alpha} = 0$

so that we find

$$(t^a)_{BC} = \left(\begin{array}{c|c} -if_{abc} \equiv (t_H^a)_{bc} & 0 \\ \hline 0 & -if_{a\alpha\beta} \equiv (t_\pi^a)_{\alpha\beta} \end{array} \right)$$

★

↗ This is the adjoint representation of H .

↖ This is the representation of H under which the Goldstone bosons transform.

Let us summarize these results in words:

■ We consider the adjoint of the group G . Under transformations of the full group, the adjoint is irreducible (if G is simple).

■ If we restrict to transformations of the subgroup H , the adjoint becomes reducible (★ has a block diagonal form)

and becomes the direct sum

$$\text{Ad}_G \stackrel{H}{\sim} \text{Ad}_H \oplus R_\pi$$

↗ adjoint of the subgroup H

↖ representation of H under which the Goldstone bosons transform

Added explanation (not in the original notes)

Why the off-diagonal blocks vanish. The upper-right block is $-if_{ab\alpha}$ and the lower-left is $-if_{a\alpha c}$; both vanish by $f_{ab\alpha} = 0$ (Lecture 25 [p. 41 *****]) together with total antisymmetry. Block-diagonality is precisely the statement that the two sets of indices — unbroken b, c and broken α, β — do not mix under H . Equivalently, the Lie algebra decomposes as $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ with $[\mathfrak{h}, \mathfrak{h}] \subset \mathfrak{h}$ and $[\mathfrak{h}, \mathfrak{m}] \subset \mathfrak{m}$: a *reductive* decomposition.

The dimensions add up. Ad_G has dimension $\dim G$; Ad_H has dimension $\dim H$; R_π therefore has dimension $\dim G - \dim H$ — the number of Goldstones, as it must. The decomposition is a statement about how the $\dim G$ generators reorganize when one only looks

at the surviving symmetry.

“Irreducible if G is simple”. The adjoint of a simple Lie algebra is irreducible precisely because a proper invariant subspace would be a proper ideal, and simple algebras have none. Once one restricts to a subalgebra, invariant subspaces can and do appear — here two of them — so the representation becomes reducible. This is the general mechanism by which a multiplet of a big symmetry splits into several multiplets of a smaller one, and it is exactly what happens physically when a symmetry is spontaneously broken.

R_π need not be irreducible. It is a genuine representation of H , but it can itself decompose further — in the Standard Model, for instance, the three would-be Goldstones of $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$ split under $U(1)_{\text{em}}$ into charges $+1, -1, 0$, matching W^+, W^-, Z . What the argument guarantees is only that the Goldstones *form* a representation of H , not that it is irreducible.

12 Example: $SO(3) \rightarrow SO(2)$ [p. 68*****]–[p. 69*****]

EXAMPLE

We consider the breaking pattern $SO(3) \rightarrow SO(2)$. We have $\dim G - \dim H = 3 - 1 = 2$ Goldstone bosons. We want to find under which representation of $SO(2)$ they transform.

First, we construct the adjoint representation of $SO(3)$. Since, locally, $SO(3) \cong SU(2)$, the Lie algebra is

$$[T^A, T^B] = i \epsilon_{ABC} T^C$$

$\rightsquigarrow$ the adjoint is defined by

$$(t^A)_{BC} = -i \epsilon_{ABC}$$

We construct the three matrices explicitly.

$$t^1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad t^2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \quad t^3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

We consider $\Phi_0 = \begin{pmatrix} 0 \\ 0 \\ v \end{pmatrix}$, so that $H = SO(2)$ is the subgroup of rotations around the third axis.

We restrict to the subgroup $H = SO(2)$, with generator

$$t^3 = \left(\begin{array}{cc|c} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

The subgroup H is abelian, thus its adjoint representation is the trivial one with vanishing generator (the structure constants are zero).

$\Downarrow$ We extract

$$t_\pi^3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

This is the generator of H in the representation of H under which the two Goldstone bosons transform.

This is the generator of $SO(2)$ in the fundamental representation acting on two-dimensional vectors. We thus find that the two Goldstones $\pi^1(x)$ and $\pi^2(x)$ transform as a vector under $H = SO(2)$.

Added explanation (not in the original notes)

Index-ordering trap flagged. On [p. 67*****] the general block form was written with the *unbroken* indices first, so that Ad_H occupied the upper-left block. In this example the natural labelling has the broken generators as 1, 2 and the unbroken one as 3, so the blocks appear in the *opposite* order: the 2×2 upper-left block is R_π and the trivial 1×1 lower-right entry is Ad_H . Both are correct; only the ordering of the basis differs, and ★ should be read as “block diagonal”, not “ Ad_H is always in the top-left corner”.

Checking the decomposition. $\text{Ad}_{SO(3)}$ is three-dimensional. Restricted to $SO(2)$ it splits as $\text{Ad}_{SO(2)} \oplus R_\pi$ with $\dim \text{Ad}_{SO(2)} = 1$ (trivial, since $SO(2)$ is abelian and all its structure constants vanish) and $\dim R_\pi = 2$. So the familiar statement “a 3-vector decomposes under rotations about the z axis into a 2-vector in the xy plane plus a scalar z -component” is exactly the representation-theoretic content here.

The physical reading. The two Goldstone bosons of $SO(3) \rightarrow SO(2)$ carry charge ± 1 under the unbroken $SO(2) \cong U(1)$ — the combinations $\pi^1 \pm i\pi^2$ are eigenvectors of t_π^3 with eigenvalues ± 1 . Any interaction one writes down must respect this, which is a real constraint: for instance, a term linear in a single π^α is forbidden, and Goldstones can only be produced in pairs of opposite charge. This is H doing its Wigner–Weyl job of organizing the spectrum.

Sanity check against Lecture 26. $SO(3) \rightarrow SO(2)$ is the $N = 3$ case of $SO(N) \rightarrow SO(N-1)$, where the $N-1$ Goldstones were found to sit in the fundamental of $SO(N-1)$ — because the potential depended on them only through $\theta^2 = \sum_a \theta_a^2$. Here $N-1 = 2$ and the fundamental of $SO(2)$ is the two-dimensional vector. The two routes agree.

PART IV — The Higgs mode, and Goldstones as coordinates

[p. 70*****]–[p. 72*****]

13 Putting the Higgs mode back in [p. 70*****]–[p. 72*****]

■ The Higgs mode

The ansatz $\Phi(x) = \exp(i\pi^\alpha(x)X^\alpha/f_\pi)\Phi_0$ does not exhaust the dynamics since we have to include the **Higgs mode**. We discuss the two examples of breaking

$$\begin{array}{llll}
 SO(N) & \longrightarrow & SO(N-1) & \\
 N \text{ d.o.f.} & & N-1 \text{ d.o.f.} & \rightsquigarrow 1 \text{ d.o.f. is missing} \\
 \Phi(x) = \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right) \Phi_0 & & & \\
 \nearrow & & \uparrow & \\
 2N \text{ d.o.f.} & & 2N-1 \text{ d.o.f.} & \rightsquigarrow 1 \text{ d.o.f. is missing} \\
 SU(N) & \longrightarrow & SU(N-1) &
 \end{array}$$

$$SO(N) \longrightarrow SO(N-1)$$

We write

$$\Phi(x) = \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right) \begin{pmatrix} 0 \\ \vdots \\ v + h(x) \end{pmatrix}$$

Remember that the generators are skew-symmetric, $(X^\alpha)^T = -X^\alpha$.

We compute

$$\begin{aligned}
 \Phi^T \Phi &= (0, \dots, v+h) \left[\exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \right]^T \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right) \begin{pmatrix} 0 \\ \vdots \\ v+h \end{pmatrix} = \\
 &= (0, \dots, v+h) \exp\left(\frac{i\pi^\alpha (X^\alpha)^T}{f_\pi}\right) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right) \begin{pmatrix} 0 \\ \vdots \\ v+h \end{pmatrix} = \\
 &= (0, \dots, v+h) \underbrace{\exp\left(\frac{-i\pi^\alpha X^\alpha}{f_\pi}\right) \exp\left(\frac{i\pi^\alpha X^\alpha}{f_\pi}\right)}_{= \mathbb{I}} \begin{pmatrix} 0 \\ \vdots \\ v+h \end{pmatrix} = \\
 &= (v+h)^2
 \end{aligned}$$

$$\rightsquigarrow \Phi^T \Phi = (v+h)^2$$

Consequently, in the broken phase, we find the potential

$$V(h) = \frac{1}{2}(2\mu^2)h^2 + \frac{\lambda}{4}h^4 + v\lambda h^3$$

h is the massive mode with $m_h^2 = 2\mu^2$.

$$SU(N) \rightarrow SU(N-1)$$

We write

$$\Phi(x) = \exp\left(\frac{i\pi^\alpha(x)X^\alpha}{f_\pi}\right) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \vdots \\ v + h(x) \end{pmatrix}$$

The generators are now hermitian, $(X^\alpha)^\dagger = X^\alpha$.

We find

$$\Phi^\dagger\Phi = \frac{1}{2}(v + h)^2$$

and, as far as the potential in the broken phase is concerned, one finds the usual result

$$V(h) = \frac{1}{2}(2\mu^2)h^2 + \frac{\lambda}{4}h^4 + v\lambda h^3$$

with Higgs mass $m_h^2 = 2\mu^2$.

Added explanation (not in the original notes)

The counting that motivates the section. The Goldstone matrix acting on the fixed vacuum produces configurations that all lie on $\mathcal{M}_0$, so it accounts for $\dim \mathcal{M}_0$ of the field degrees of freedom. But the theory has more: N real fields in the $SO(N)$ case and $2N$ in the $SU(N)$ case. Since $\dim S^{N-1} = N - 1$ and $\dim S^{2N-1} = 2N - 1$, exactly one degree of freedom is unaccounted for in each — the *radial* direction, off the manifold. Promoting $v \rightarrow v + h(x)$ restores it. In effect the field is now written in generalized polar coordinates: an angular part (the Goldstone matrix, tangent to $\mathcal{M}_0$) times a radial part ($v + h$, normal to it).

Why the exponentials cancel. For $SO(N)$ the broken generators, being generators of a real orthogonal group, are antisymmetric and purely imaginary, so iX^α is real antisymmetric and $(iX^\alpha)^T = -iX^\alpha$; the transpose of the exponential is therefore its inverse, and $\Phi^T\Phi$ collapses. For $SU(N)$ the generators are hermitian, so the exponential is unitary and $\Phi^\dagger\Phi$ collapses in the same way. In both cases the statement is simply that a group element preserves the invariant, which is what defines the group. This is the exact generalization of $|e^{i\pi/v}|^2 = 1$ in Lecture 24 [p. 17*****].

Everything depends only on h . The potential is a function of the invariant alone, and the invariant is a function of h alone — so V is the same one-variable Mexican-hat function met in Lecture 24, with the same mass $m_h^2 = 2\mu^2$ and the same cubic and quartic self-couplings, *independently of N and of whether the group is SO or SU* . That is a striking universality, and it is the reason the massive mode is always called “the Higgs” regardless of the breaking pattern.

Where the Goldstones went. They have not disappeared — they are absent from V but

very much present in $\mathcal{L}_{\text{KIN}}$, where the derivatives acting on the Goldstone matrix generate the whole non-linear sigma model, including the h - π interactions. Lecture 24 [p. 18*****]–[p. 19*****] did this computation explicitly in the abelian case; the general version is the CCWZ construction.

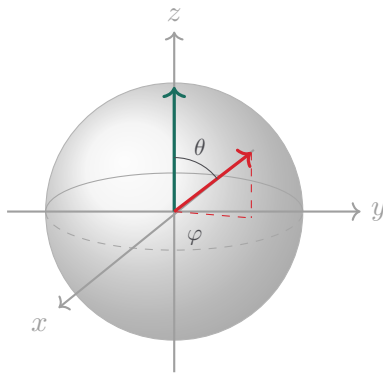
14 Goldstone bosons as local coordinates on the vacuum manifold

[p. 72*****]

■ Goldstone bosons as local coordinates on the vacuum manifold

We explore in more detail the connection between Goldstone bosons and geometry.

We start from some group theory considerations and, to make the discussion less abstract, we focus on the specific case of $SO(3) \rightarrow SO(2)$.



The vacuum manifold is

$$\frac{SO(3)}{SO(2)} \cong S^2$$

We take as broken generators J^1 and J^2 , and J^3 is unbroken.

We start from a simple identity: a generic ...

[the scan ends here]

Added explanation (not in the original notes) — Where this is heading

The scan breaks off at the words “a generic ...”. The identity being set up is almost certainly the statement that a generic $SO(3)$ rotation can be factorized into a rotation generated by the broken generators times one generated by the unbroken generator,

$$R = \underbrace{e^{i(\pi^1 J^1 + \pi^2 J^2)/f_\pi}}_{\text{coset representative}} \underbrace{e^{i\alpha J^3}}_{\in H},$$

which is the group-theoretic content of $\mathcal{M}_0 \cong G/H$ made concrete. Acting on the north pole $\Phi_0 = (0, 0, v)^T$, the H factor does nothing, and the coset representative carries Φ_0 to the point of S^2 with polar angles determined by π^1, π^2 .

Explicitly, for the two-sphere one finds that the combination $\sqrt{(\pi^1)^2 + (\pi^2)^2}/f_\pi$ is the polar angle θ and the ratio π^2/π^1 fixes the azimuth φ — so the two Goldstone fields are, quite literally, a coordinate chart on the sphere, drawn in the figure above. Small fluctuations around the vacuum correspond to points near the north pole, where the chart is well behaved; this is the precise sense of the word *local*, since a single exponential chart cannot cover all of S^2 (the south pole is at $\theta = \pi$, where the map degenerates, exactly as polar coordinates do). The payoff, which the following lecture presumably develops, is the kinetic term. Substituting

the Goldstone matrix into $\mathcal{L}_{\text{KIN}}$ produces

$$\mathcal{L} = \frac{f_\pi^2}{2} g_{\alpha\beta}(\pi) \partial_\mu \pi^\alpha \partial^\mu \pi^\beta + \cdots,$$

where $g_{\alpha\beta}$ is the G -invariant metric on G/H — for $SO(3)/SO(2)$, the round metric on the sphere, $d\theta^2 + \sin^2 \theta d\varphi^2$. This is the *non-linear sigma model*. Every feature established in this lecture then follows from geometry: the shift symmetry is the invariance of the metric under the isometries of G ; the linear action of H is the isotropy action at the chosen point; and the absence of a potential is the fact that all points of G/H have the same energy. The Goldstone bosons are maps from spacetime into the vacuum manifold, and their low-energy Lagrangian is fixed by the geometry of that manifold alone.

Summary of Lecture 27 — the logic in nine lines

Added explanation (not in the original notes) — Summary

1. For $SO(3) \rightarrow SO(2)$ the vacuum manifold is the two-sphere, its points are related by $SO(3)$, and $\Phi_0 = (0, 0, v)^T$ is left invariant by rotations about the third axis. Formally, $\mathcal{M}_0 \cong G/H$ — and since H is not normal, G/H is a manifold, not a group. [p. 53*****]
2. The coset space is built by partitioning G into left cosets $H, g_1H, g_2H, \dots$; the cosets are disjoint, cover G , and each is a copy of H . [p. 54*****]–[p. 55*****]
3. The identification with $\mathcal{M}_0$: every element of a coset gH sends Φ_0 to the *same* vacuum, because $H\Phi_0 = \Phi_0$. One coset $\leftrightarrow$ one point of $\mathcal{M}_0$, and the map is bijective. [p. 55*****]–[p. 57*****]
4. Counting parameters: $\dim G$ to specify $g \in G$, of which $\dim H$ are redundant, so $\dim \mathcal{M}_0 = \dim G - \dim H = \# \text{Goldstones}$. This closes comment 3 of Lecture 26. [p. 57*****]–[p. 58*****]
5. The general exponential parametrization: $\Phi = \exp(i\pi^A T^A / f_\pi) \Phi_0$. Since the exponential is a group element and Φ_0 minimizes V , one gets $V(\pi) = V_{\min}$ identically — the Goldstones vanish from the potential. [p. 59*****]–[p. 60*****]
6. The unbroken directions are redundant ($T^a \Phi_0 = 0$ collapses the series), so one restricts to broken generators: $\Phi = \exp(i\pi^\alpha X^\alpha / f_\pi) \Phi_0$, the *Goldstone matrix*. [p. 60*****]–[p. 61*****]
7. Under a *broken* transformation, expanding to first order in both θ and π gives $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$: a non-linear shift, which forbids every non-derivative term. [p. 61*****]–[p. 62*****]
8. Under an *unbroken* transformation, inserting $\mathbb{K} = e^{i\alpha T} e^{-i\alpha T}$ turns the action into a conjugation of the Goldstone matrix; using $[T^a, X^\alpha] = if_{a\alpha\beta} X^\beta$ (the T piece vanishes since $f_{ab\alpha} = 0$) gives $\pi^\alpha \rightarrow [e^{-i\alpha_a t_\pi^a}]_{\alpha\beta} \pi^\beta$ with $(t_\pi^a)_{\alpha\beta} = -if_{a\alpha\beta}$: linear, as befits a Wigner–Weyl symmetry. [p. 63*****]–[p. 65*****]
9. Restricted to H , the adjoint of G becomes block diagonal: $\text{Ad}_G \overset{H}{\sim} \text{Ad}_H \oplus R_\pi$, with R_π the representation carrying the Goldstones. For $SO(3) \rightarrow SO(2)$, R_π is the two-dimensional vector of $SO(2)$. Finally, restoring $v \rightarrow v + h$ gives $\Phi^T \Phi = (v + h)^2$ and the universal $V(h) = \frac{1}{2}(2\mu^2)h^2 + v\lambda h^3 + \frac{\lambda}{4}h^4$. [p. 66*****]–[p. 72*****]

Added explanation (not in the original notes) — Where this is going

The chapter has now assembled a complete picture of a spontaneously broken *global* symmetry: the vacuum manifold is G/H ; the Goldstone bosons are coordinates on it, exactly $\dim G - \dim H$ of them; they shift non-linearly under the broken generators and rotate linearly under H ; and one extra radial mode, the Higgs, carries the only non-trivial potential.

Two things remain. The immediate one is the kinetic term — writing $\mathcal{L}_{\text{KIN}}$ in terms of the Goldstone matrix produces the non-linear sigma model, whose form is dictated by the invariant metric on G/H . That is what the unfinished section on [p. 72*****] is building towards.

The larger one is gauging. Every Goldstone here is a physical massless particle, and nature does not obviously contain them. Make the symmetry local and the picture changes completely: the Goldstone fields become the parameters of the gauge transformation and can be removed by a choice of gauge, reappearing as the longitudinal polarizations of the gauge

bosons, which thereby become massive — one massive vector per broken generator. The transformation laws derived in this lecture are precisely what makes that possible: because $\pi^\alpha \rightarrow \pi^\alpha - f_\pi \theta_\alpha$ is a pure shift, a local $\theta_\alpha(x)$ can be chosen to set $\pi^\alpha(x) = 0$ everywhere. That gauge is called unitary gauge, and the mechanism is the Higgs mechanism.